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The AI Dilemma: Growth versus Existential Risk

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1 Big picture

- **Question.** How should society trade off the growth benefits of advanced AI against existential risk?
- **Core tension.** AI may dramatically raise living standards by accelerating innovation, but the same powerful AI systems may create a risk of human extinction.
- **Main economic object.** The paper studies the optimal intensity of AI use when each extra period of AI use raises consumption growth but also increases existential risk.
- **Key sufficient statistic.** In the simple model, the relevant comparison is the value of a year of life, expressed as a multiple of consumption, against the AI benefit-cost ratio g/δ .
- **Main result 1.** With log utility, the model can tolerate very large existential risks if AI delivers huge consumption gains. For example, with $g = 10\%$ and $\delta = 1\%$, it is optimal to let AI run for 40 years, producing a 55-fold consumption gain but a one-third extinction probability.
- **Main result 2.** With CRRA utility and $\gamma \geq 2$, the conclusion changes sharply. Bounded utility and rapidly declining marginal utility make the planner much more conservative.
- **Main result 3.** A singularity with infinite consumption does not automatically dominate when utility is bounded. Infinite consumption gives finite flow utility when $\gamma > 1$.
- **Main result 4.** Mortality improvements are different from consumption improvements. Longer life expectancy is in the same units as existential risk, so AI-driven health improvements can justify substantially higher risk cutoffs.
- **Main result 5.** Near-zero social discounting makes society less willing to risk an infinite future when AI only raises consumption. But if AI also improves longevity, even very future-oriented planners may tolerate sizable existential risk.
- **Economic takeaway.** The answer depends less on a single estimate of AI risk and more on deep normative parameters: curvature of utility, value of life, social discounting, and whether AI extends life as well as consumption.

2 Data, calibration, and measurement

2.1 No empirical treatment; calibrated thought experiments

- The paper is a quantitative theory paper rather than an empirical causal study.
- It uses simple models to discipline the tradeoff between AI-enabled growth and extinction risk.
- The numerical exercises are meant to clarify economic forces, not to forecast exact AI risk.

Interpretation: The paper's value comes from making the tradeoff explicit. It shows which assumptions make AI use look acceptable versus unacceptable.

2.2 Core calibration values

- Initial consumption is normalized to $c_0 = 1$.
- AI-enabled fast growth is set to $g = 10\%$ per year in the simple model.
- The paper considers flow existential-risk rates of $\delta = 1\%$ and $\delta = 2\%$ in the simple model.
- In the richer model, baseline growth without AI is $g_0 = 2\%$.
- Baseline mortality is $m_0 = 1\%$, corresponding to life expectancy of about 100 years.

- The mortality-improvement case assumes AI cuts mortality to $m_{ai} = 0.5\%$, corresponding to life expectancy of about 200 years.
- The private discount rate and the baseline social discount adjustment are set around 1% in the main calibrations.

Interpretation: The numbers are intentionally stark: AI is given a large possible benefit, while existential risk is kept as a transparent parameter.

2.3 Value of life calibration

- The value of a statistical life year in the United States is taken to be roughly six times annual consumption.
- The paper therefore sets

$$v(c_0) = 6,$$

where $v(c)$ is the value of one year of life in consumption units, divided by consumption itself.

- The utility intercept $\bar{u}$ is chosen separately for each γ to match $v(c_0) = 6$.

Interpretation: This makes the model's value of life consistent with observed willingness to pay to reduce mortality risk, while preserving the comparison across risk-aversion parameters.

3 Models and equations

3.1 Simple static model

AI is used for T periods. It raises consumption to

$$c = c_0 e^{gT}, \tag{1}$$

but survival is

$$S(T) = e^{-\delta T}. \tag{2}$$

With a constant population N and discount rate ρ , expected social welfare is

$$EU(T) = e^{-\delta T} \frac{N}{\rho} u(c_0 e^{gT}). \tag{3}$$

Interpretation: Increasing T gives more growth and more extinction risk. The tradeoff is one-dimensional.

3.2 First-order condition

The planner should continue using AI as long as the marginal growth benefit exceeds the marginal extinction-risk cost:

$$\delta \cdot \frac{N}{\rho} u(c) \leq g \cdot \frac{N}{\rho} u'(c)c. \tag{4}$$

After canceling common terms, define

$$v(c) \equiv \frac{u(c)}{u'(c)c}. \tag{5}$$

For an interior optimum,

$$v(c^*) = \frac{g}{\delta}. \tag{1}$$

Interpretation: The planner runs AI until the value of a year of life reaches the AI benefit-cost ratio g/δ . If the current value of life already exceeds g/δ , the planner shuts AI down immediately.

3.3 CRRA utility and the value of life

The paper uses CRRA utility with a constant chosen so that life is worth living relative to death:

$$u(c) = \begin{cases} \bar{u} + \log c, & \gamma = 1, \\ \bar{u} + \frac{c^{1-\gamma}}{1-\gamma}, & \gamma \neq 1. \end{cases} \quad (6)$$

The implied value of life is

$$v(c) = \begin{cases} \bar{u} + \log c, & \gamma = 1, \\ \bar{u}c^{\gamma-1} + \frac{1}{1-\gamma}, & \gamma \neq 1. \end{cases} \quad (2)$$

Interpretation: With log utility, $v(c)$ rises only with $\log c$. With $\gamma > 1$, utility is bounded and marginal utility falls quickly, so $v(c)$ rises much faster with consumption.

3.4 Richer dynamic model

The richer model lets future generations matter explicitly. Social welfare is

$$W = \int_0^\infty e^{-\rho_s t} N_t U_t dt, \quad (7)$$

where

$$N_t = N_0 e^{bt} \quad (8)$$

and the lifetime utility of a cohort born at date t is

$$U_t = \int_0^\infty e^{-(\rho+m)a} u(c_{t+a}) da. \quad (9)$$

Interpretation: The model separates individual discounting ρ , social discounting ρ_s , birth growth b , mortality m , and consumption growth g .

3.5 Closed-form social welfare

With CRRA utility and growth rate g , the paper obtains

$$U_t = \frac{\bar{u}}{\rho + m} + \frac{c_t^{1-\gamma}}{1-\gamma} \cdot \frac{1}{\rho + m + (\gamma - 1)g}. \quad (10)$$

Substituting into social welfare gives

$$W(g, m) = \frac{N_0 \bar{u}}{(\rho + m)(\rho_s - b)} \quad (11)$$

$$+ \frac{N_0 c_0^{1-\gamma}}{1-\gamma} \cdot \frac{1}{(\rho + m + (\gamma - 1)g)(\rho_s - b + (\gamma - 1)g)}. \quad (3)$$

Interpretation: This expression makes it possible to compare baseline growth and mortality to AI growth and mortality.

3.6 Existential-risk cutoff

Without AI, the economy has (g_0, m_0) . With AI, it has (g_{ai}, m_{ai}) but faces one-time existential risk δ . AI is optimal if

$$W(g_0, m_0) < (1 - \delta)W(g_{ai}, m_{ai}). \quad (12)$$

The critical cutoff is

$$\delta^* =$$

$$\frac{W(g_{ai}, m_{ai}) - W(g_0, m_0)}{W(g_{ai}, m_{ai})}. \quad (13)$$

Interpretation: If actual AI extinction risk is below δ^* , AI use is optimal in the model. If actual risk is above δ^* , the planner should not use AI.

3.7 Singularity case

When $\gamma > 1$, a singularity with infinite consumption does not imply infinite utility. Flow utility is bounded by $\bar{u}$, so social welfare with infinite consumption is

$$W_{sing} = \frac{N_0 \bar{u}}{(\rho + m_{ai})(\rho_s - b)}. \quad (14)$$

Interpretation: This is the central reason why a singularity is not infinitely valuable under bounded utility. Infinite consumption merely reaches the utility upper bound.

3.8 Near-zero social discounting and life expectancy

When the social discount rate approaches zero, the paper shows that mortality improvements can keep existential-risk cutoffs high. If AI doubles life expectancy, a simple expression is

$$\delta^* = \frac{1/2}{\rho \cdot LE + 1}. \quad (15)$$

With $\rho = 0.01$ and $LE = 100$, this gives

$$\delta^* = \frac{1/2}{2} = \frac{1}{4}. \quad (16)$$

Interpretation: Even a very future-oriented planner may tolerate sizable existential risk if AI also gives future generations much longer lives.

4 Conceptual design and interpretation

4.1 Why the utility curvature matters so much

- Log utility values large consumption gains strongly because utility is unbounded and declines slowly in marginal utility.
- CRRA utility with $\gamma > 1$ has bounded utility, so extra consumption eventually has little welfare value.
- Since existential risk removes all future utility, bounded consumption utility makes the planner more cautious.

Interpretation: The same growth and risk numbers can imply radically different policy conclusions depending on γ .

4.2 Why mortality improvements are special

- Consumption gains pass through diminishing marginal utility.
- Mortality improvements expand the duration of life and therefore operate in the same dimension as existential risk.
- If AI gives people more years of life when it succeeds, this partly offsets the risk that no future life exists when it fails.

Interpretation: AI health innovations can change the welfare calculus even when pure consumption gains do not.

4.3 What the paper does not try to do

- It does not estimate the true probability of AI catastrophe.
- It does not model governance institutions, enforcement, or international coordination.
- It does not model AI safety investment explicitly.
- It abstracts from distributional conflicts, non-utilitarian ethical views, and irreversibility.

Interpretation: The model is not a final policy prescription. It is a framework for understanding which parameters and ethical assumptions drive the answer.

5 Tables and figures

This paper has two original figures and three original tables. The visuals below are directly cropped from the uploaded paper and grouped compactly to improve reading speed.

5.1 Figure 1 and Table 1: utility curvature and the simple model

Read: Figure 1:

- For $\gamma > 1$, CRRA flow utility is bounded above by $\bar{u}$.
- The marginal utility of consumption falls sharply.
- This makes extremely high consumption less valuable than in log utility.

Read: Table 1:

- With $\delta = 1\%$ and log utility, the model chooses $T^* = 40$, consumption rises to $c^* = 54.60$, and total existential risk is 0.33.
- With $\delta = 1\%$ and $\gamma = 2$, the model chooses only $T^* = 4.5$ and total risk is 0.04.
- With $\delta = 1\%$ and $\gamma = 3$, the model chooses $T^* = 2.4$ and total risk is 0.02.
- With $\delta = 2\%$, the model shuts AI down immediately for all three γ values.

Economic message: Utility curvature is decisive. Log utility tolerates a large extinction gamble for huge consumption gains; $\gamma \geq 2$ makes the planner much more conservative.

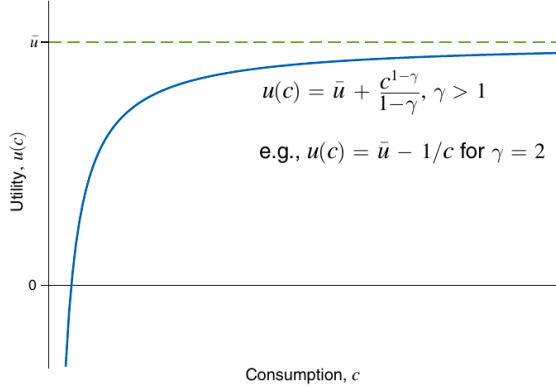


FIGURE 1. BOUNDED FLOW UTILITY WHEN $\gamma > 1$

(a) Bounded flow utility when $\gamma > 1$.

TABLE 1—CONSUMPTION AND EXISTENTIAL RISK: SIMPLE MODEL

γ	$\delta = 1\%$			$\delta = 2\%$		
	c^*	T^*	Existential risk	c^*	T^*	Existential risk
1	54.60	40.0	0.33	1	0	0
2	1.57	4.5	0.04	1	0	0
3	1.27	2.4	0.02	1	0	0

(b) Simple-model quantitative results.

5.2 Figure 2 and Table 2: value of life, mortality improvements, and singularities

Read: Figure 2:

- The value of life is normalized to six at current average US consumption.
- With log utility, the value of life rises slowly with consumption.
- With $\gamma = 2$ or $\gamma = 3$, the value of life rises much more quickly.

Read: Table 2:

- With near-log utility, existential-risk cutoffs are very high: 0.540 for fast growth and 0.916 for a singularity when mortality does not improve.
- With $\gamma = 2$, the fast-growth cutoff without mortality improvements falls to 0.022.
- With $\gamma = 3$, the corresponding cutoff falls to 0.005.
- Mortality improvements change the result sharply: with $\gamma = 2$, cutting mortality from 1% to 0.5% raises the fast-growth cutoff from 0.022 to 0.267.

Economic message: Consumption gains face diminishing marginal utility, but longevity gains are directly comparable to existential risk. That is why mortality improvements can rescue otherwise risky AI use.

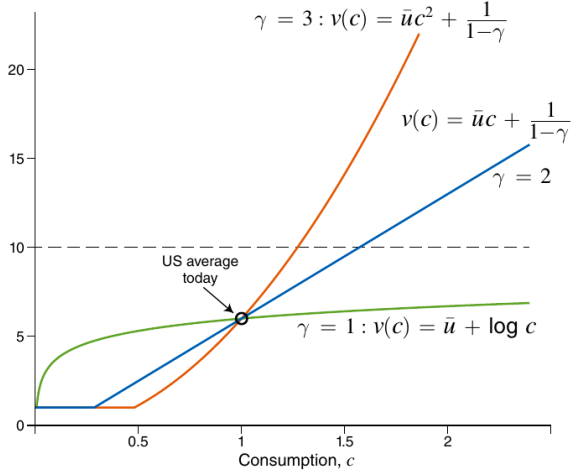


FIGURE 2. THE VALUE OF A YEAR OF LIFE, $v(c)$

(a) Value of one year of life.

TABLE 2—EXISTENTIAL RISK CUTOFFS: MORTALITY IMPROVEMENTS AND SINGULARITIES

γ	Fast growth: $g_{ai} = 10\%$		Singularity: $g_{ai} = \infty$	
	m_{ai}		m_{ai}	
	1%	0.5%	1%	0.5%
1.01	0.540	0.678	0.916	0.937
2	0.022	0.267	0.024	0.268
3	0.005	0.254	0.005	0.254

(b) Risk cutoffs with mortality improvements and singularities.

5.3 Table 3: near-zero social discounting

Read:

- Table 3 lowers the social discount rate from $\rho_s = 1\%$ to $\rho_s = 0.05\%$.
- Without mortality improvements, the existential-risk cutoff becomes much smaller for $\gamma = 2$ and $\gamma = 3$.
- With mortality improvements, cutoffs remain high: about 0.251 for $\gamma = 2$ and 0.250 for $\gamma = 3$.
- The table shows that future-oriented social discounting strengthens caution about risking future generations, unless the AI also makes those future generations live longer.

Economic message: Near-zero social discounting is not automatically pro-AI or anti-AI. It lowers risk tolerance for pure consumption gains, but it can preserve high risk tolerance when AI improves mortality.

TABLE 3—EXISTENTIAL RISK CUTOFFS WITH NEAR-ZERO SOCIAL DISCOUNTING

γ	Baseline $\rho_s = 1\%$		Near-zero social discounting $\rho_s = 0.05\%$	
	m_{ai}		m_{ai}	
	1%	0.5%	1%	0.5%
1.01	0.540	0.678	0.525	0.646
2	0.022	0.267	0.002	0.251
3	0.005	0.254	0.000	0.250

Table 3: Existential-risk cutoffs with near-zero social discounting.

6 Short summary

- The paper studies when AI-enabled growth is worth the associated existential risk.
- In the simple model, AI use raises consumption by $c_0 e^{gT}$ and survival falls by $e^{-\delta T}$.

- The key first-order condition is $v(c^*) = g/\delta$.
- The value of life $v(c)$ depends strongly on utility curvature.
- With log utility, large consumption gains can justify substantial extinction risk.
- With CRRA utility and $\gamma \geq 2$, bounded utility makes the planner much more conservative.
- A singularity with infinite consumption is not infinitely valuable when utility is bounded.
- Mortality improvements are powerful because they directly offset mortality and existential-risk losses.
- With $\gamma = 2$, mortality improvements can raise the risk cutoff from around 2.2% to around 26.7%.
- Near-zero social discounting reduces risk tolerance for pure consumption gains but does not necessarily do so when AI extends life expectancy.
- The paper does not claim to estimate the true AI risk; it clarifies the welfare logic.

7 Conclusion

7.1 Core conclusion

Jones shows that the optimal response to transformative AI is extremely sensitive to utility curvature, the value of life, social discounting, and whether AI affects mortality. Pure consumption growth can justify large AI risks under log utility, but much less so under bounded CRRA utility. If AI also lengthens life, however, substantially higher existential-risk cutoffs may be rational.

7.2 Economic intuition

The core intuition is that consumption and life are not the same margin. Consumption gains become less valuable as consumption rises, especially with $\gamma > 1$. Existential risk destroys all future life. Mortality improvements extend life and therefore operate on the same welfare margin as extinction risk. This is why health-related AI benefits can matter more than even spectacular consumption growth.

7.3 Takeaway

The AI dilemma is not solved by one number for AI growth or one number for AI risk. The conclusion depends on how society values consumption, life, future generations, and mortality improvements. The paper's main lesson is to make these assumptions explicit before drawing strong policy conclusions about whether to accelerate, delay, or stop advanced AI.